

WEEK SIX - THE ART OF DESCRIBING

Thursday, 6 March 2025, 12:45 - 2:15 PM EST.

M-Lab, Harvard Art Museums

THE NETHERLANDS in the time of Rembrandt and Vermeer was also a time of Dutch scientific achievement. A new type of science involved *seeing* and *describing* the natural world. Galileo Galilei (1564-1642) was one leader who used telescopes to see and describe the night sky. But being in Florence, Galileo was an exception. Most contemporary scientists who found new ways to use optics to see and describe the world were in Northern Europe. The Danish Tycho Brahe (1546-1601) built the largest observatory in Europe. The German Johannes Kepler (1571-1630) improved Galileo's telescope and mathematical understanding of the solar system. The Dutch Antony von Leeuwenhoek (1632-1723) developed a microscope that enabled him to discover microorganisms and bacterial life.

When a new world is seen – whether with telescope or microscope – it has to be described. Quantitative and reliable visual description – *pictorial art* – became necessary for scientific communication, as important as the written word or equation.

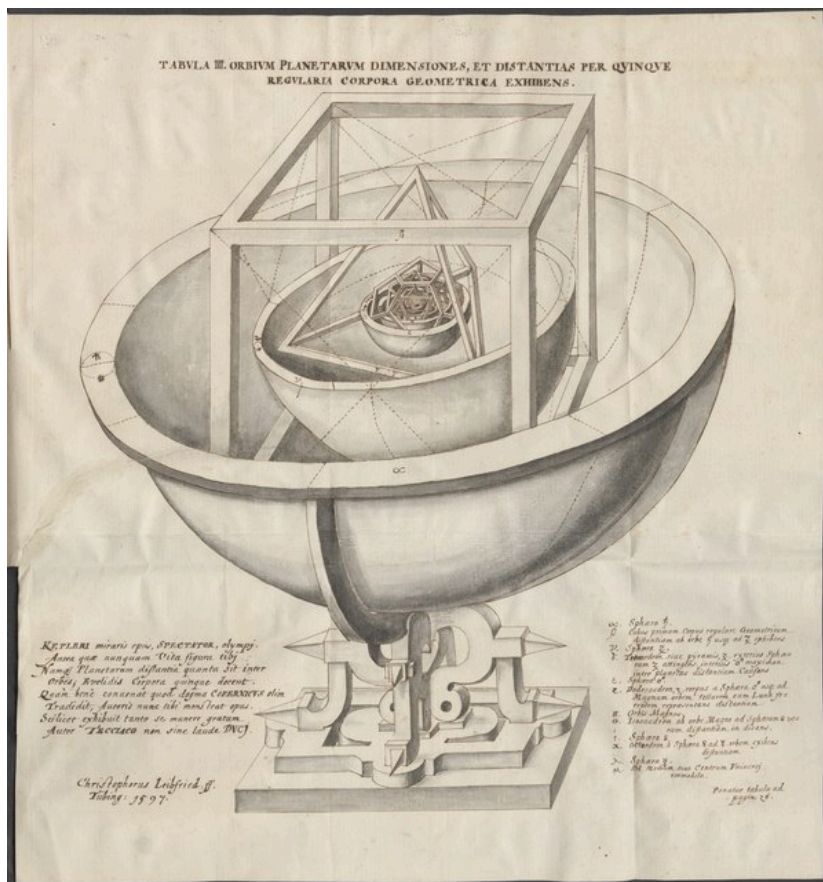


Figure 134: Platonic solid model of the Solar System from *Mysterium Cosmographicum* by Kepler.

NORTHERN EUROPE was especially receptive to optical science. The writings of Sir Constantijn Huygens, Lord of Zuilichem, (1596-1687), provides one window into contemporary Dutch attitudes (Fig. 135). Constantijn was a polymath. He corresponded with Descartes, championed Rembrandt, and inspired his son to become a mathematical physicist. Christiaan Huygens (1629-1695) would discover 'Huygens Principle' of wave propagation.

Constantijn had the opportunity to use a camera obscura that was bought by his friend Cornelis Drebbel, another leading scientist. His reaction reveals the impact of this new image-making device and the importance of visual over written communication.

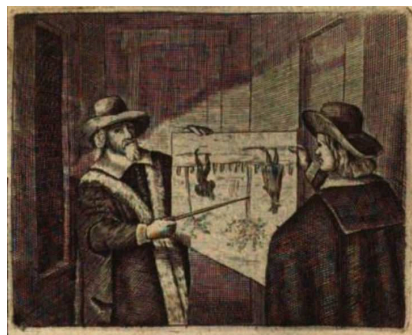
It is not possible to describe for you the beauty of [the camera obscura] in words: all painting is dead by comparison, for here is life itself, or something more noble, if only it did not lack words.

A POEM BY CONSTANTIJN is similarly revealing. Seeing is both a means to knowledge and way to experience God's creation:

O you who give the eyes and the power,
Give eyes through this power:
Eyes once made watchful, Which see the totality of all there is to see.

LENSES AND MIRRORS in the 17th century were not perfect. Images were distorted. The circumspet worried whether optics revealed truth or lies. Optical devices were validated in 1604 when Kepler discovered that the human eye literally *works* like a camera obscura. The eye, like the camera obscura, is equipped with lens and pinhole that projects images onto the back surface. Kepler wrote: "ut pictura, ita vision" or "sight is like a picture".

IN A DUTCH HANDBOOK BY VAN BEVERWYCK, *Wercken der Genees-Konste* from 1667, two men illustrate the operation of the eye by standing in a dark room with a pinhole fitted by a lens. The outside world is cast onto a surface, just like an image is cast onto the back of the eye. In the camera obscura, pictures are painted with rays of light, not with brush and pigment.



Updated: March 4, 2025



Figure 135: *Portrait of Constantijn Huygens and his Clerk* by Thomas de Keyser. Huygens's table is the story of his duties, interests, and talents – a musical instrument, architectural plans, and terrestrial and celestial globes. [Link to painting at the National Gallery in London](#)

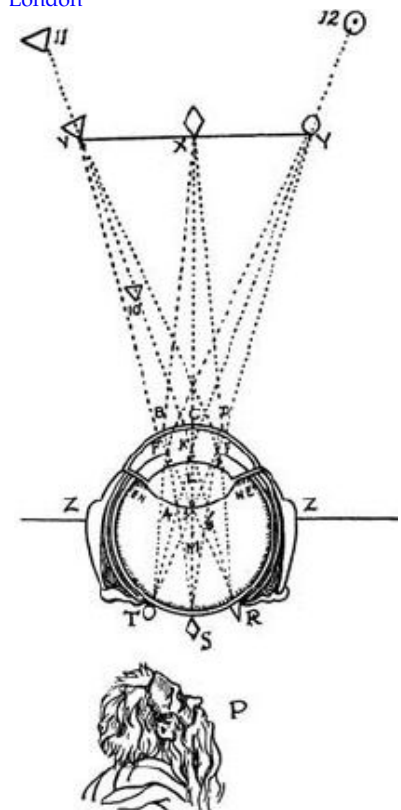


Figure 136: Illustration of the theory of the retinal image from *La Dioptrique* by Descartes.

NORTHERN EUROPEAN ART was long known for detailed, precise, and patiently crafted descriptions of the natural world. When Constantijn looked into Drebbel's microscope, he found another visual worlds that needed to be described. These descriptions required new art.

CONSTANTIJN HUYGENS became a supporter of Leeuwenhoek (1632-1723), a contemporary of Vermeer (1632-1675) in the small town of Delft. Leeuwenhoek discovered microorganism by inventing the most powerful single-lens microscope in Europe. His microscope had magnification and clarity far beyond what Drebbel would have shown to Constantijn. Leeuwenhoek made the first drawings of bacteria, which he called *animalcules*.

In the 17th century, learning to draw was as essential to a complete education as learning languages or mathematics. As a student, Constantijn had wanted to study drawing with the Dutch realist Jacob de Gheyn II (1565-1629). De Gheyn's representations of flora and fauna were famed for scientific precision (Fig. 138). Such talents were needed for the new world that Constantijn and Leeuwenhoek saw in the microscope. Constantijn wrote:

For in fact, this concerns a new theater of nature, another world, and if our revered predecessor De Gheyn had been allotted a longer life span, I believe he would have advanced to the point to which I have begun to push people (not against their will); namely to portray the most minute objects and insects with a finer pencil, and then to compile these drawings into a book to be given the title of the *New World* .

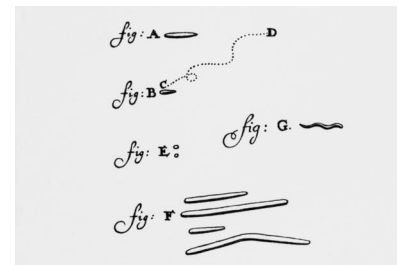


Figure 137: Animalcules from a letter by Leeuwenhoek from 1683

Figure 138: Studies of a Fantastic Bird, Toad, Frog, and Dragonfly by de Gheyn, 1596-1602. [Link to Morgan Library and Museum.](#)

FRANCIS BACON (1561-1626) – philosopher at Cambridge University and the Court of Elizabeth I – has been called the inventor of the modern scientific method. He advocated *empiricism* – knowledge by inductive reasoning and carefully seeing and describing Nature. Constantijn Huygens was influenced by Bacon, as were Robert Hooke (1635-1703) and Isaac Newton (1643-1727).

Robert Hooke corresponded with Leeuwenhoek and made his own microscopes. Huygens had advocated a scientific program of accurate visual representation of the microscopic world. This program was realized by Hooke when he published *Micrographia* in 1665.⁸ Hooke used his microscopes to discover and describe the surprisingly exquisite architectures of insects.

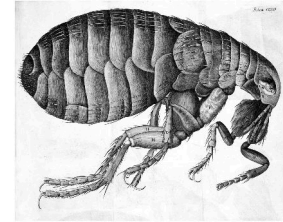


Figure 139: Flea from *Micrographia* by Robert Hooke, 1665

⁸ Martin Kemp. Hooke's housefly. *Nature*, 393(6687):745-745, 1998. ISSN 0028-0836

MICROGRAPHIA uses both words and images to describe Nature. Comparing Hooke's words with drawings reveals the weakness of language in describing the new worlds he saw in the microscope.

The Eye of a Fly in one kind of light appears almost like a lattice, drill'd through with abundance of small holes... in the Sunshine they look like a surface cover'd with golden Nails' in another posture, like a surface cover'd with pyramids; in another with Cones; and in other postures of quite other shapes.

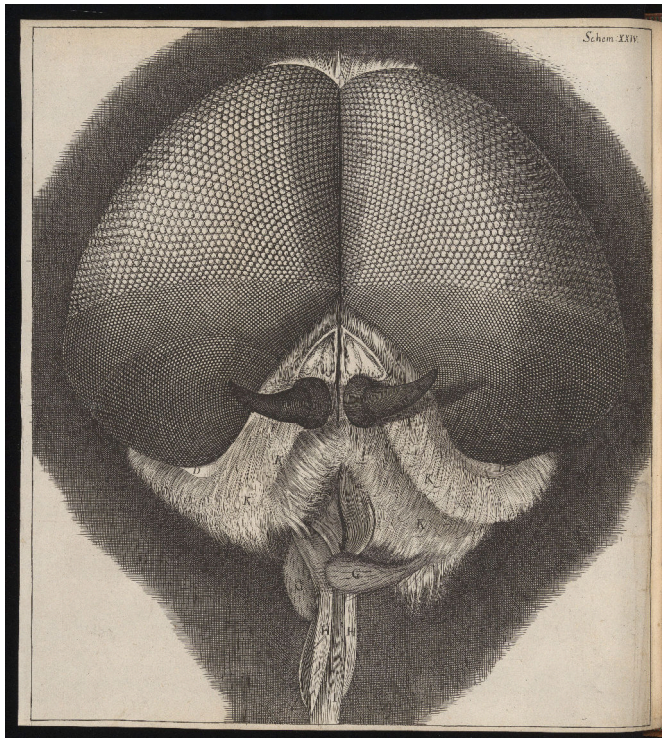


Figure 140: Eyes and head of a grey drone fly by Robert Hooke, 1665. [Link to Wellcome Collection.](#)

COMMON THEMES IN ART AND SCIENCE reveal shared cultural attitudes and tastes. When de Gheyn painted a sick mouse, he captured the details of its swollen eyes and fur standing on end from multiple angles. These drawings reflect the artist's devotion to photographic realism, but also a scientific process of thought as the artist contemplates the mouse, rolling it over in his mind.



Figure 141: *Four studies of a diseased mouse* by Jacques de Gheyn. [Link to Rijksmuseum.](#)

THE DUTCH STILL LIFE also reveals a taste for microscopic dissection and displaying every facet of an object. Specialists in the still life, like Pieter Claesz (1597-1661) and Willem Kalf (1619-1693), would paint tableau of cheese, fish, fruit, and serving vessels, revealing inside, outside, and underside. Foods are sliced open. Lemon rinds are unwound. Objects are dissected and exposed. The eye is drawn to every optical detail that is faithfully captured with paint and brush.



Figure 142: *Still Life* by Pieter Claesz, 1627. Still life did not develop into its own distinct style until the 17th century. Pieter Claesz in Haarlem specialized in these still lifes, devising different arrangements to display all dimensions of common objects. [Link to Timken Museum.](#)



Figure 144: *Still Life with Tazza* by Pieter Claesz, 1636. Various objects strewn on a table, including sliced lemon. The tazza is the lavishly decorated silver drinking vessel. [Link to Mauritshuis](#)

DAVID HOCKNEY came to alternative optical conclusion with a still life by Juan van der Hamen y León (Fig. 147).⁹ Here, grapes, sliced melon, and sliced pomegranates are painted separately. None would stay fresh by the time that another was painted. This work looks like a montage of separately painted objects, each captured in a snapshot moment of time much briefer than painting requires. An optical tool would have let the artist work quickly, before the fruit rotted.

⁹ David Hockney. *Secret Knowledge*. Viking Studio, New York, 2006. ISBN 978-0-14-200512-5



Figure 145: *Still Life with Fruit and Glassware* by Juan van der Hamen y León, 1626. [The Museum of Fine Arts, Houston](#)

KEPLER first described the retinal image as a “picture”. What the eye sees is effectively painted by rays of light on the back of the eye. This model of the eye has an underlying statement about visual truth and the validity of the camera obscura for depicting the world. Kepler likely owned a “tent camera obscura”, which he used for landscapes. The image formed by such a tent camera obscura would look like an image made by a modern camera.

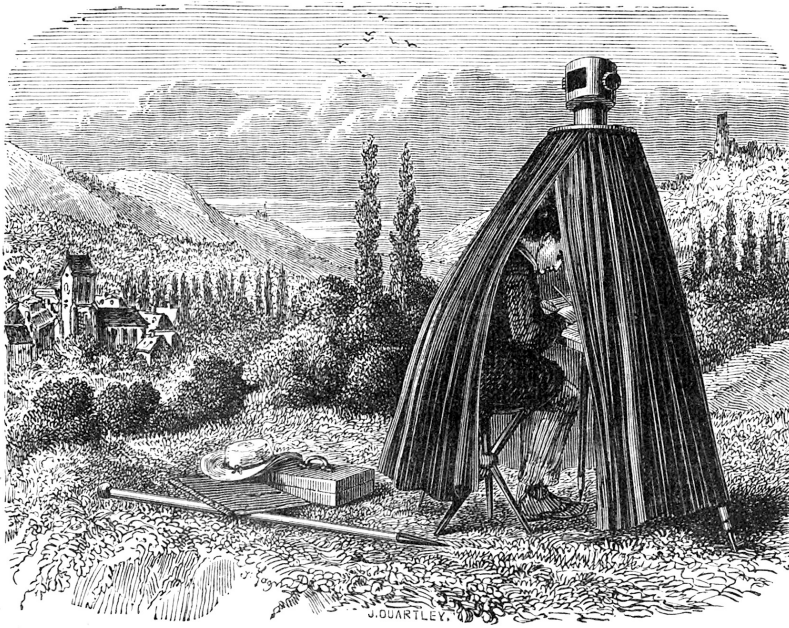


Figure 146: *Tent Camera Obscura* from an 1858 book on physics.

VERMEER'S *View of Delft* has the uncanny look of a color photograph. Vermeer had found a new approach to landscape painting. Delft has no obvious geometric design or narrative center. Instead, Vermeer captured a glimpse of a quiet city on a quiet morning. How did Vermeer see Delft in a way that looks like a color photograph?

A CAMERA OBSCURA might have helped Vermeer to see differently. Vermeer might have admired the unique way that a camera obscura created an instant and complete glimpse of the world. He then might have tried to replicate its visual effects, the same effects that occur in modern color photography, in his own painting.



Figure 147: *View of Delft* by Vermeer
[Link to painting.](#)

VERMEER'S *Milkmaid* also has the narrative effect of color photography. Vermeer captures a glimpse of a domestic scene, a quiet moment as the milkmaid pours. Linear perspective does not create a three-dimensional stage for a drama. Sunlight illuminates an everyday scene from life, which Vermeer transfers to paint.

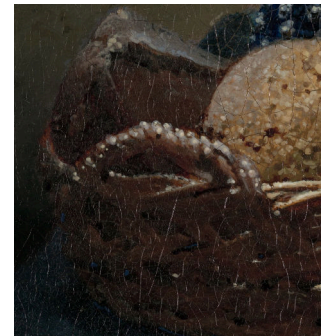
CERTAIN OPTICAL EFFECTS appear in Vermeer paintings that are unique to lens-based device. In both *The Milkmaid* and *The View of Delft*, Vermeer dabs 'circles of confusion', bright globules of coalesced highlights that simulate an out-of-focus effect caused by diffracting and reflecting light. Some circles of confusion are out of place, like from the non-reflective matte surface of the bread basket in *The Milkmaid*. Vermeer does not faithfully copy what he sees. He sprinkles optical effects from a camera obscura like ornaments.

ANOTHER OPTICAL EFFECT in *The Milkmaid* is optical focus. Objects have sharp focus in an image plane and soft focus in other planes. Optical focus only happens when an image is made with a lens. The naked eye automatically and unconsciously focuses to every point in the real world. You cannot see variable focus unless you look at a flat image that was made by a fixed lens. Vermeer might have liked the focus of the camera obscura, and mimicked its effect in a painting.



So long as that woman from the
Rijksmuseum
in painted quiet and concentration
keeps pouring milk day after day
from the pitcher to the bowl
the World hasn't earned
the world's end.

—Wisława Szymborska



ANOTHER VIEW OF DELFT is by Carel Fabritius (1622-1654). Fabritius has been called the most brilliant of Rembrandt's students. Ten years older and living in Delft, he might have taught Vermeer. Little remains of his work. He was killed by the Delft gunpowder explosion on 12 October 1654 that destroyed his studio and paintings. Only a dozen paintings survive, including *The Goldfinch* and *View of Delft*.

Fabritius's *View of Delft* has a wide-angle perspective. It resembles the panoramic photograph when a smartphone is swept across a field of view. Now, Fabritius's painting is hung and seen on a flat surface in London. But the painting was probably originally mounted on a curved surface in a rounded box.

Maybe the *View of Delft* is an act of imagination. The artist predicted, in his mind's eye, what a future smartphone might do. Another possibility is that Fabritius used a device, like a camera obscura with a curved surface. Such machines that mapped scenes onto cylindrical surfaces did exist (Fig. 150). A third possibility is that Fabritius copied what he saw through a lens. A wide-angle camera shot of the same scene in modern Delft has a similar optical distortion as Fabritius's painting (Fig. 149).

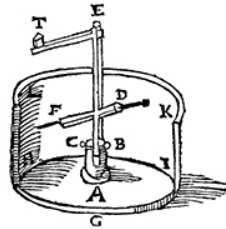


Figure 150: Baldassare Lanci device to project a view on the inside of a cylinder (Kemp, 1990)

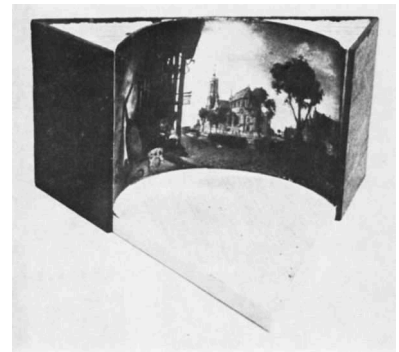


Figure 148: Interior view of a reconstruction of a perspective box that might have contained Fabritius's *View of Delft*.



Figure 149: View of the church painted by Fabritius, but photographed through a double convex lens. From Wheelock (1973).



Figure 151: *The View of Delft* by Fabritius, 1652. This very small painting (15x30 cm) is a view of the Nieuwe Kerk in Delft. Perspective is emphasized by two musical instruments, typical virtuoso subjects for foreshortening. The violin recedes into picture space, as if very close to the viewer. The church loom forward, and the street bends as if curved by a lens. [Link to National Gallery](#).

DELFT WAS DIFFERENT from what Fabritius saw in 1652 or Vermeer saw in 1661. In 1654, Delft was shattered by a gunpowder explosion, equivalent to 22 tons of TNT. This explosion destroyed a quarter of the city and killed hundreds including Fabritius. The explosion and its aftermath were painted by Egbert van der Poel. Vermeer's *View of Delft* cannot be a faithful visual image. In painting after painting, Vermeer leaves out grim reality in a city afflicted by war and plague. Vermeer chooses stillness. He chooses quietude.



Figure 152: *A View of Delft after the Explosion of 1654* by Egbert van der Poel, 1654, [Link to National Gallery](#).



Figure 153: *A View of Delft During the Explosion of 1654* by Egbert van der Poel, 1654, [Link to Rijksmuseum](#).

THE CAMERA OBSCURA became a scientific and commercial device in the 17th century. But its basic optical principle, what happens when an image is cast by a pinhole in a darkened room, had been known for centuries. Before Kepler, Leonardo da Vinci had wondered whether the camera obscura was a model for the human eye. Da Vinci wanted to understand visual truth. He believed that the painter should depict what is projected into the eye:

The painter's mind should be like a mirror, which transforms itself into the color of the thing that it has as its object, and is filled with as many likenesses as there are things placed before it. Therefore, painter, knowing that you cannot be good, if you are not a versatile master in reproducing through your art all the kinds of forms that nature produces – which you will not know how to do if you do not see and represent them in your mind.

THE FLOWERS, ROCKS, AND LANDSCAPE of his *Madonna of the Rocks* are painted with photographic realism, as da Vinci follows his own advice. Unlike Vermeer, da Vinci also takes liberties with what he sees, assembling objects to follow a painterly design. In *View of Delft* and other paintings, the painter and any painterly design seems to disappear. The image tells the whole story.



Figure 154: *Madonna of the Rocks* by Leonardo da Vinci, 1483-1486

WRITING

Ernst Gombrich is the best known art historian of the 20th century. His *The Story of Art* is the foundation of innumerable courses in Art History. Gombrich also taught Svetlana Alpers. Svetlana Alpers wrote her landmark *The Art of Describing*, about the special optical qualities of Northern European painting.

Download and read the Introduction to *The Art of Describing* (more if you want) and Gombrich's review of her book.

Comment on Alpers, and whether you think Gombrich agrees with her.



PAINTING

- Nard Kwast, Dutch reality TV star known for his recreation of Rembrandt's *Nightwatch* in the Rijksmuseum will lead a workshop to paint a Dutch Still Life. [Link to slides](#)

READING

- Introduction of *The Art of Describing* [Download entire book](#)
- Review of *The Art of Describing* [Download paper](#)

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